Software Requirements Specifications

UniVerseX

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INTRODUCTION AND BACKGROUND

UniVerseX is a mobile application designed to create a social platform for university students to communicate, share knowledge, and conduct peer-to-peer transactions. Currently, students' discussions scatter across WhatsApp, Facebook, and Instagram, making important information difficult to retrieve and maintain continuity. Our platform consolidates academic discussions, social interaction, and student commerce into a single, structured environment accessible on both iOS and Android devices.

1.1. Problem Statement

University students face fragmented communication channels where academic discussions, social posts, and marketplace activity are scattered across multiple unstructured platforms. This fragmentation results in:

- Loss of important academic information in group chats
- Difficulty finding answers to repeated questions
- Unsafe, unmoderated student commerce
- No dedicated space for campus-specific discussions

UniVerseX solves this by providing a unified platform where students can:

- Post and discuss course-related topics in moderated communities
- Share confessions and social content anonymously
- Buy, sell, and barter textbooks, notes, and campus accessories
- Access all information within a single, organized interface in various dedicated communities

1.2. Background

In universities, communication is essential for academic success and campus life. Students currently rely on general-purpose social media platforms not specifically designed for academic purposes. WhatsApp groups become cluttered, Facebook groups are public and unprofessional, and Instagram stories fade within 24 hours. There is a clear demand for a dedicated student platform that is community-driven.

The proposed solution draws inspiration from successful platforms: **Reddit, Piazza** and **Empor** remaining lightweight and prototype-focused using **MIT App Inventor**. The platform will emphasize moderation, user safety, and organized architecture to ensure a healthy discussion environment.

1.3. Scope

UniVerseX will be developed as a cross-platform mobile application (iOS/Android) with the following core features:

Authentication & Profile:

- Email/password registration and login
- User profile with username and reputation score
- Profile view showing user's posts and marketplace listings

Discussion Features:

- Create text-based posts with optional image uploads
- Post categorization (Question, Discussion, Confession)
- Moderated communities
- Comment threads under posts
- Upvote/downvote mechanism for post visibility
- Search posts by keyword, category, or community (Limited)

Marketplace Features:

- List items for sale or barter (books, notes, accessories, etc.)
- Item details with images, price, and barter options
- Contact seller through in-app messaging
- User ratings for marketplace transactions

Moderation:

- Community moderators (admins) appointed per community
- Ability to report inappropriate content
- Admin dashboard for content removal
- Spam and abuse prevention

Technical Stack:

- Frontend: MIT App Inventor (cross-platform blocks)
- Backend: Firebase (Realtime Database + Authentication)
- Image Storage: Cloudinary (free tier)
- Target: Android + iOS

1.4. Objectives

- Provide a centralized, organized platform for student communication and knowledge sharing
- Eliminate information fragmentation across multiple apps
- Create safe, moderated spaces for academic and social discussions
- Enable peer-to-peer student commerce with built-in trust mechanisms
- Improve student engagement and sense of campus community
- Demonstrate practical application of software engineering principles in a real-world project

1.5. Challenges

Real-time Synchronization: Ensuring feed updates reflect new posts and comments instantly across users

Database Complexity: Managing relationships between users, posts, comments, and marketplace listings

Image Handling: Optimizing image uploads and storage within app size constraints

User Trust: Building confidence in marketplace transactions (ratings, reviews, dispute resolution)

Performance: Managing and organizing a large number of posts and comments

MIT App Inventor Limitations: Lack of advanced features like *nested* queries and complex business logic

1.6. Learning Outcomes

Upon completion, team members will have gained:

- Understanding of *Software Requirements Specification* documentation
- Mobile app development using MIT App Inventor and visual programming
- Cloud database design and real-time synchronization with Firebase
- User authentication and security best practices
- Marketplace and community platform architecture
- Team collaboration and version control (document management)
- Understanding user interaction and *system design*
- ***Software engineering lifecycle*** from specification to prototype

1.7. Nature of End Product

UniVerseX will be delivered as a **functional prototype** demonstrating core functionality:

- Cross-Platform app
- Fully functional discussion and post features
- Working marketplace with image uploads
- Real-time Firebase backend integration
- Moderated community system
- User authentication and profile management

The prototype prioritizes functionality over polish; advanced features like performance optimization and complex analytics are **excluded**.

1.8. Completeness Criteria

The project will be considered complete when:

1. Users can register, log in, and maintain profiles
2. Posts can be created, displayed, and deleted by authors
3. Comments appear under posts in real-time
4. Upvote/downvote system functions correctly and reflects in post scoring
5. Communities can be created and moderated
6. Marketplace listings can be posted with images and descriptions
7. Search functionality returns relevant results
8. Feed displays posts sorted by newest and popularity
9. Marketplace transactions show contact options

1.9. Business Goals

- Increase student engagement within the university ecosystem
- Provide a safe alternative to unmoderated third-party platforms
- Reduce dependency on external social media for campus discourse
- Enable sustainable student commerce with built-in buyer protection
- Build a foundation for potential university adoption and expansion

1.10. Related Works

Reddit (reddit.com): Community-driven discussion platform with upvote/downvote ranking, moderated subreddits, and user reputation. UniVerseX adopts the moderated community model and post-based discussion format.

Piazza (piazza.com): Academic Q&A platform used in universities with instructor monitoring and structured answers. UniVerseX incorporates the educational focus and moderation hierarchy.

OLX / Facebook Marketplace: Peer-to-peer marketplace platforms. UniVerseX integrates marketplace functionality with trust ratings and in-app messaging.

UniVerseX differentiates itself by combining all three elements (*discussion, academic focus, and marketplace*) in one platform specifically designed for universities, rather than adapting general-purpose tools.

1.11. Document Conventions

Key Terms: User, Post, Comment, Community, Moderator, Listing, Marketplace are used consistently throughout
Terminology: "Post" refers to discussion content; "Listing" refers to marketplace items
Hierarchy: Communities contain Posts; Posts contain Comments; Marketplace is a separate feature area
Numbering: Sections follow **IEEE SRS** standard numbering format

OVERALL DESCRIPTION

2.1. Product Features

The major features of the system are listed below:

A. Discussion & Posts

- Create posts with title, body text, tags and optional images
- **Tags**: Question, Discussion, Confession, etc. to improve organization and searchability
- Upvote/downvote mechanism with score calculation
- Comment threads on each post
- Edit and delete own posts
- Pin important posts (moderators only)
- Search posts by keyword, tag, or community

B. Communities

- Create communities around topics, courses, or interests
- Join communities
- View community-specific feed
- Moderator appointment and role management
- Community rules and description
- Community moderation tools (pin, remove posts)

C. Marketplace

- List items for sale or barter with images
- Item categories (Books, Notes, Accessories, Electronics, etc.)
- Seller contact information and ratings
- In-app messaging between buyer and seller
- Transaction status tracking (Available, Sold, Exchanged)
- User reputation/ratings system

D. User & Authentication

- Email/password registration and login (**must be university mail**)
- Profile customization (username, bio, profile picture)
- View user's posts and marketplace listings
- Reputation score (based on upvotes and transaction ratings)
- Follow users (optional, for future phases)

E. Moderation & Safety

- Report inappropriate content
- Moderator dashboard per community
- Content removal and user warnings
- Spam detection and prevention
- Transaction dispute resolution tools

2.2. User Classes and Characteristics

1. Regular Students (Primary Users)

- University students aged 18 – 25
- Basic to intermediate mobile app experience
- Use the app for academic discussions, social posts, and buying/selling items
- Expect fast, intuitive interface with minimal learning curve
- May create 5 – 10 posts per week on average

2. Moderators (Secondary)

- Experienced users appointed by admins or community creators
- *Responsibility*: monitor content, remove spam, enforce community guidelines
- Higher privileges: ability to pin/remove posts, warn users
- Typically 1 – 2 per community of 100+ members

3. Administrators (System Admins)

- *Italics* indicate special role; appointed by development team
- Full platform access: view all posts, remove any content, manage moderators
- Ability to disable accounts and enforce platform-wide policies

4. Guest Users (Optional, Future Phase)

- Non-registered users who can view public posts (read-only)
- Cannot create posts, comment, or access marketplace
- Encouraged to register for full experience

2.3. Operating Environment

Platform Specifications:

- **Mobile OS:** Android 8.0+ and iOS 8.0+
- **Hardware:** Minimum 2GB RAM, 50MB storage
- **Network:** Requires active internet connection (WiFi or mobile data)

Development & Deployment:

- **Development Tool:** MIT App Inventor (visual blocks programming)
- **Backend:** Firebase (Realtime Database + Authentication + Hosting)

- **Image Storage:** Cloudinary
- **Testing:** Android emulator + physical device testing

Browser Requirements:

- Modern browsers supporting Service Workers
- **HTTPS** required for all communications

2.4. Design and Implementation Constraints

MIT App Inventor Limitations:

- Limited to visual block programming (no direct code)
- No native support for complex nested queries
- Maximum app size ~50MB after compilation
- Limited offline capability

Firebase Free Tier:

- 100 concurrent connections maximum
- 1GB total storage
- Rate limiting on reads/writes
- No built-in full-text search (*mitigation*: indexed keywords)

Image Handling:

- Cloudinary free tier limited to 10GB/month bandwidth
- Images must be optimized before upload
- *Base64 encoding* in MIT is memory-intensive

Cross-Platform Support:

- iOS support via Progressive Web App (not native)
- Native Android recommended for full feature parity
- Testing required on both platforms despite code reuse

Development Scope:

- Project prototype status; *not production-ready*
- Scalability features (caching, CDN, load balancing) excluded
- Advanced analytics and machine learning not included
- Real-time notifications may have delays

2.5 Assumptions and Dependencies

Assumptions:

- All users have access to mobile devices and have internet connectivity
- Users possess basic mobile app literacy
- University emails are available for authentication
- Firebase services remain available throughout project
- Cloudinary API availability for image uploads

Dependencies:

- **MIT App Inventor**: Platform for development and compilation
- **Firebase**: Realtime Database, Authentication, optional Cloud Functions
- **Cloudinary API**: Image upload
- **Google Account:** Required to access Firebase and MIT App Inventor
- **Internet Services**: Continuous connectivity for real-time sync

External Risks:

- Dependency on Firebase availability and pricing changes
- MIT App Inventor updates may affect compilation
- Cloudinary service disruptions